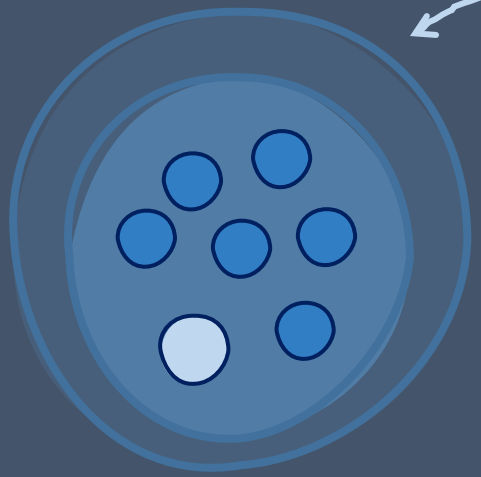
Effects of Classroom Heterogeneity on Students‘ Psycho-Social Outcomes – A Systematic Review

EARLI

Graz, Austria

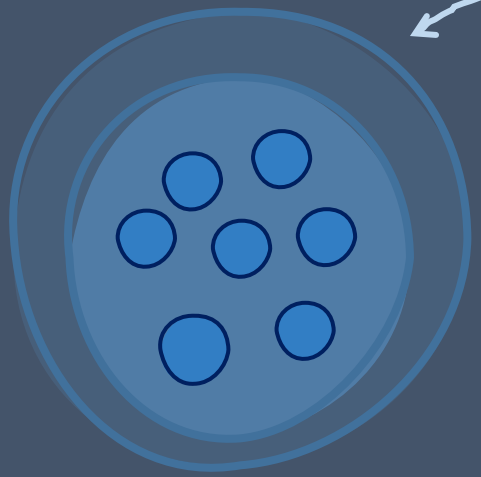
27.08.2025

RAHMENPROGRAMM EMPIRISCHE BILDUNGSFORSCHUNG
GEFÖRDERT VOM
Bundesministerium für Bildung und Forschung

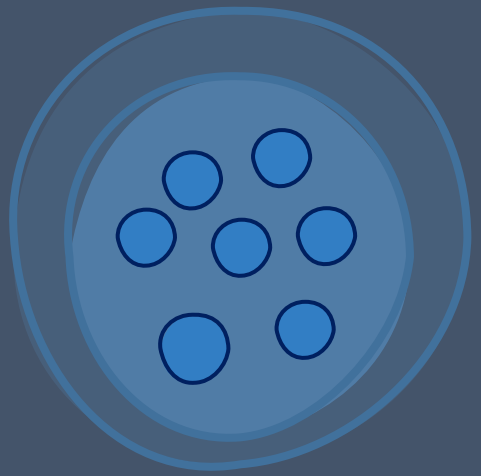


School contexts are central to students‘ intellectual and social-psychological development.

(Eccles & Roeser, 2011)



Classroom Composition Research



Editorial in Journal of Cognitive Education and Psychology | Classroom Composition Research Volume 15, Issue 2

An Overview of Classroom Composition Research on Social-Emotional Outcomes: Introduction to the Special Issue

Müller, Christoph M. | Zurbriggen, Carmen L. A.

Journal of Cognitive Education and PsychologyClassroom Composition Research Vol 15 Issue 2, Jan 2016, DOI: 10.1891/1945-8959.15.2.163

“[T]here appears to be a growing discontent with only considering the classroom averages in predicting social-emotional outcomes”

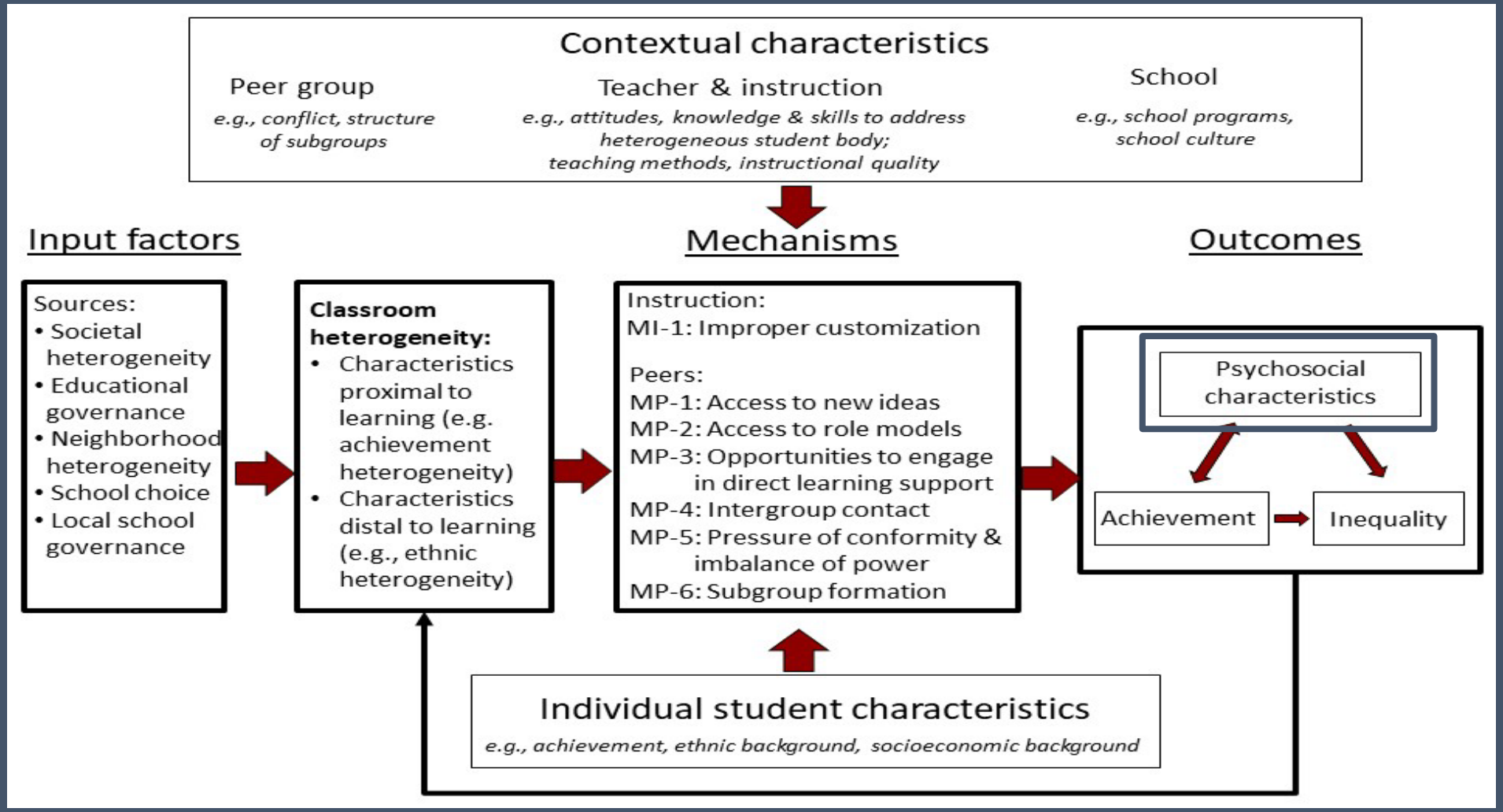
→ *Classroom heterogeneity*

as a facet of classroom composition

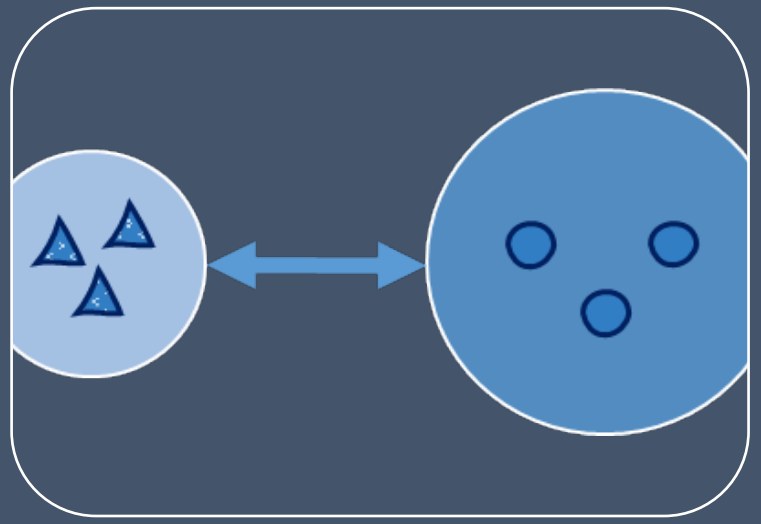


"[T]here appears to be a growing discontent with only considering the classroom averages in predicting social-emotional outcomes"

The Classroom Heterogeneity Effects in Education (CHEE-) Framework (Rjosk et al., submitted)



Prior integrative research: Focus on ethnic heterogeneity



Intergroup Outcomes

Thijs & Verkuyten (2014), Brown & Juvonen (2018)



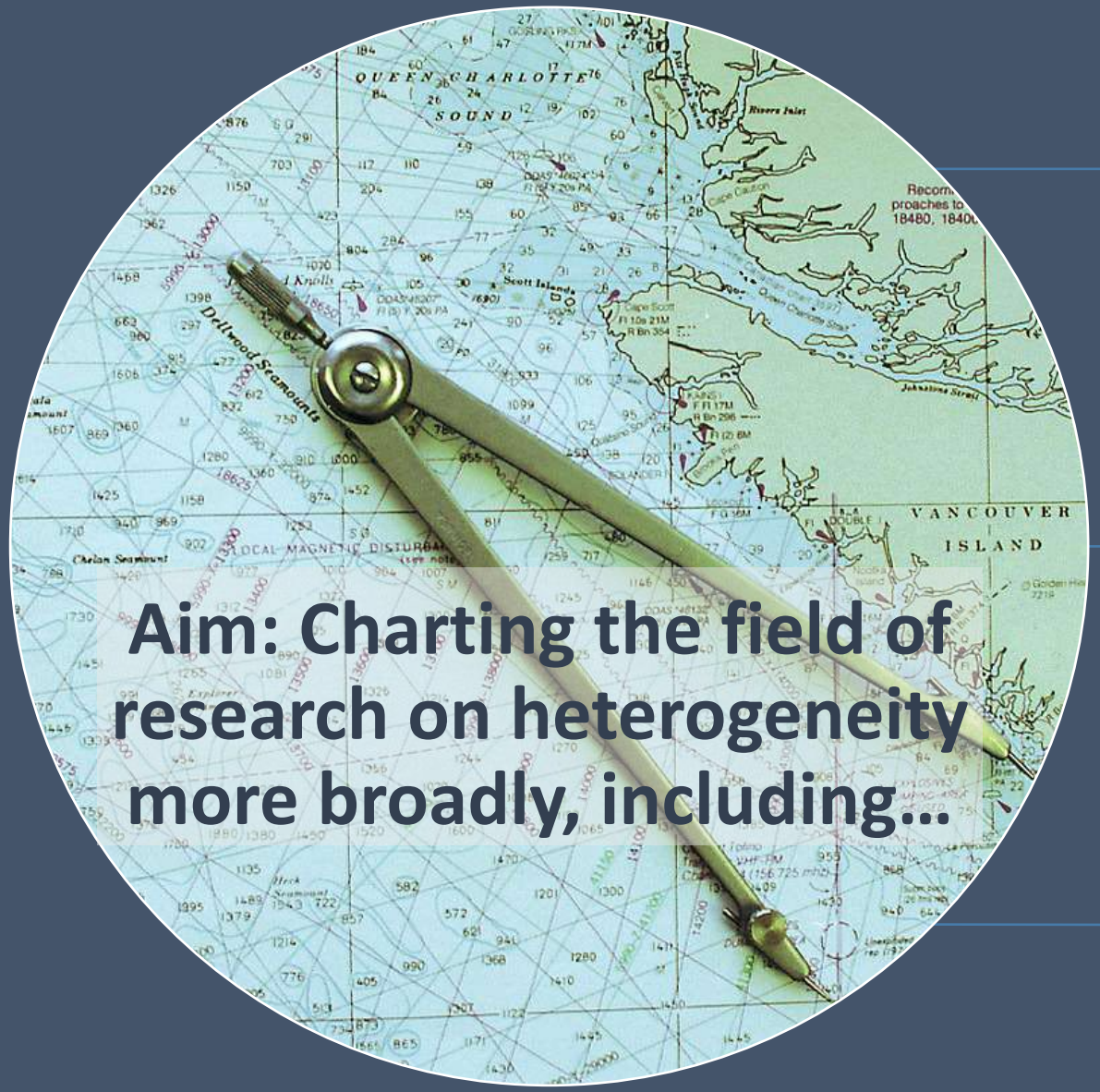
Social-emotional outcomes

Yip et al. (2019), Rucinski (2022)



Victimization and bullying

Azeredo et al. (2015), Basilici et al. (2022)



Aim: Charting the field of research on heterogeneity more broadly, including…

multiple dimensions of heterogeneity

multiple disciplines

broad psycho-social outcome domains

Method – Systematic Review: Eligibility

- Research data bases: PsychInfo, ERIC and Scopus
- Publication between 1990 – 2024
- Sample: primary or secondary school
- Multivariate quantitative methods

IV: Heterogeneity in

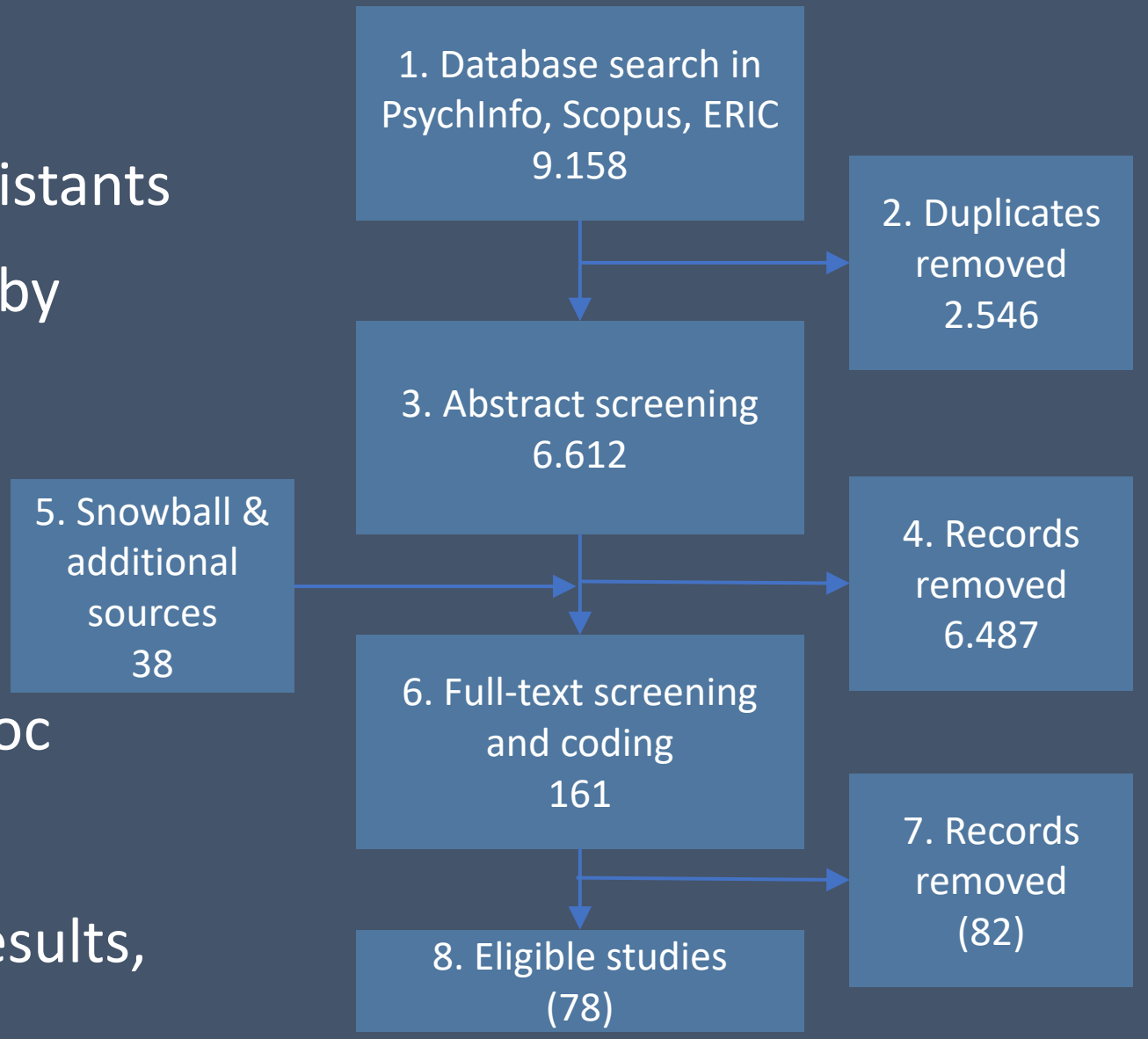
- Achievement,
- Ethnic background, Language background,
- Social / socioeconomic background,
- Gender,
- Special educational needs or other

DV: psycho-social characteristics and social relationships:

- social behavior (e.g., victimization, violence, social support, help, cooperation)
- feeling of connection (e.g., attachment, belonging, classroom climate)
- contact (e.g., friendship, peer relations, segregation, inter-ethnic contact)
- social-cognitive outcomes (e.g., social skills, pro-social competence, tolerance, trust)

Method – Systematic Review

- Abstract screening
  - 15% double coded by research assistants
  - Insecurities and conflicts resolved by discussion
- Full-text screening and coding
  - double coding for all articles
  - tandem with at least 1 doc/post-doc
  - Coding of information on
    - Sample, Variables, Methods, Results, Limitations



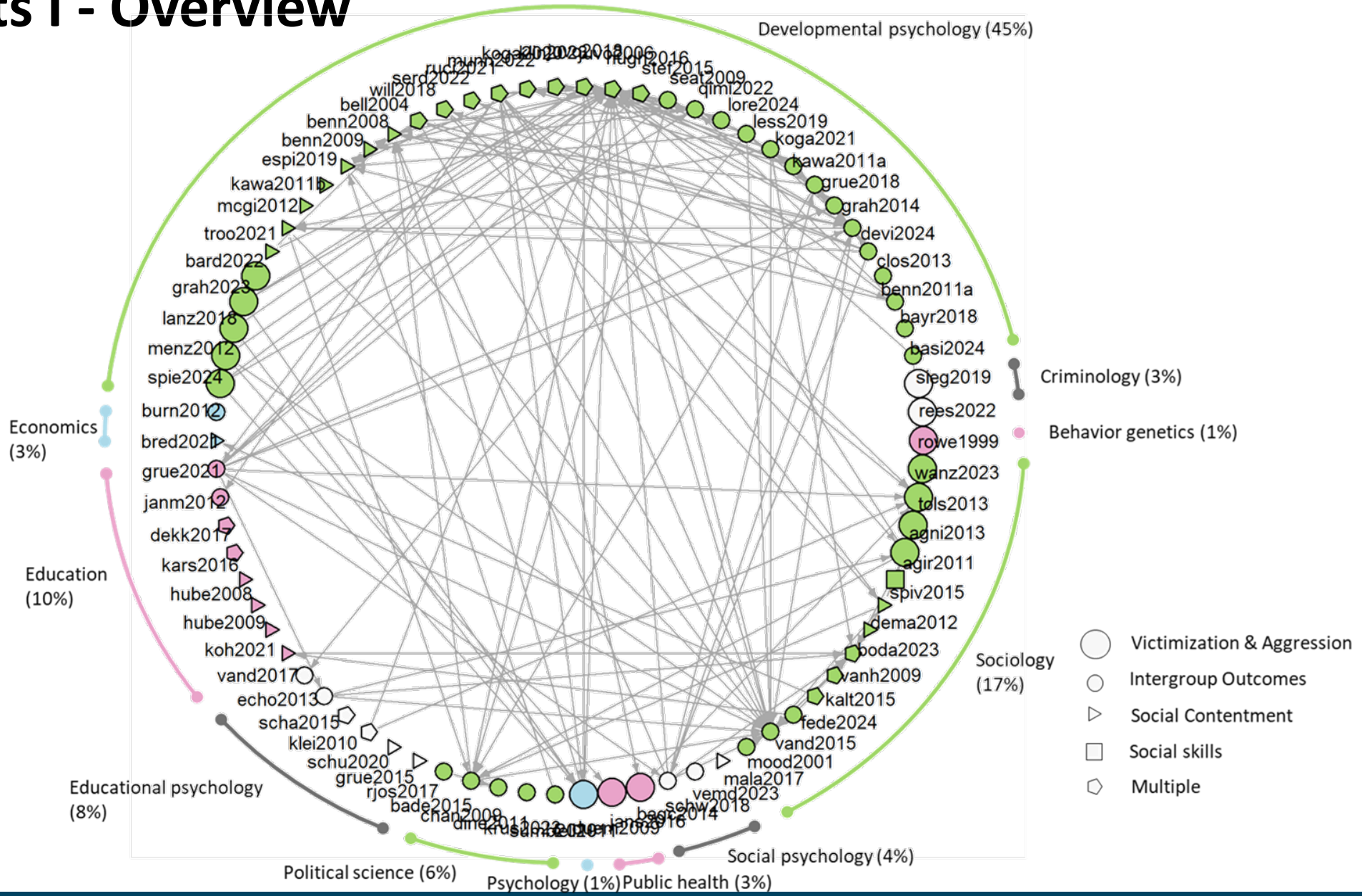
Study Descriptives

Regions	N	%
Europe	38	49
USA	33	42
Africa	1	1
Multiple	7	9

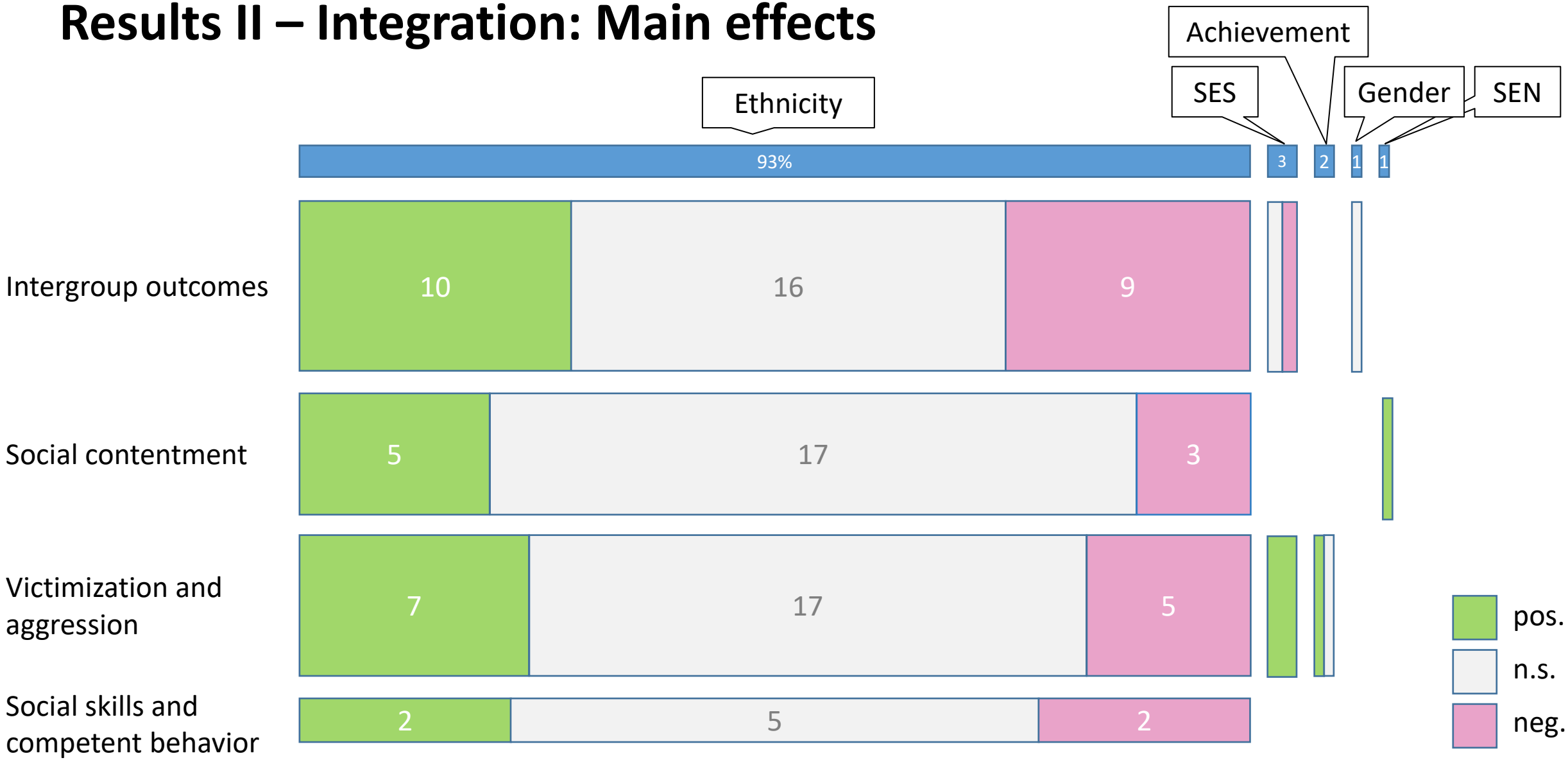
Outcome domains	N	%
Intergroup Outcomes	35	45
Social-emotional outcomes	27	35
Victimization and Aggression	25	32
Skills and prosocial behavior	5	6
Multiple	16	21

Dimensions	N	%
Ethnicity	69	86
SES	5	6
Achievement/IQ	3	4
Gender	1	1
Special educational needs	1	1
Multiple/other	1	1

Results I - Overview



Results II – Integration: Main effects



Spotlight: Victimization

Theoretical bases:

- Imbalance of Power (Olweus, 1992)



Protective Heterogeneity



Erosion of norms



- Social Disorganization Theory (Shaw & McKay, 1942)

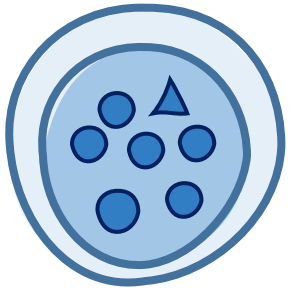
Spotlight: Victimization

Negative (5)	Negative (2)	Null (7)	Positive (4)
Graham & Echols (2023), Juvonen et al. (2018), Lanza et al. (2018) Graham et al. (2014) Juvonen et al. (2006)	Felix & You (2011), Spiegler et al. (2024)	Fedeli & Triventi (2024), Klein & Cornell (2010), Munniksma et al. (2022), Bardach et al. (2022) (+ some smaller studies: Agirdag et al., 2011; Schultze-Krumbholz et al., 2020; Wan et al., 2024, Tolsma et al., 2013; Serdiouk et al., 2022)	Jansen et al. (2016), Rees & Winfree (2022) Agnich & Miyazaki (2013) Due et al. (2009)











Spotlight: Victimization

California (highly het.) sample	Diverse studies show no effects	Low heterogeneity samples Different het. dimensions

Recommendations:

- Explore sub-group effects
- Look at mechanisms

Summary

- The effects of heterogeneity on psycho-social outcomes…
  - … are studied in diverse disciplines (psychology, education, sociology, economics, political science)
  - … are mostly confined to effects of ethnic diversity
  - … do examine a variety of psycho-social outcomes

- The evidence available to-date…
  - is often inconclusive about the effect of ethnic heterogeneity
  - is insufficient in other heterogeneity dimensions (language, SES, achievement, gender), but first results seem to point to negative effects of heterogeneity on psycho-social outcomes

Future directions

- Future research should…
  - identify moderators
  - use more different and larger data sets

  - consider additional heterogeneity dimensions
  - consider multidimensional heterogeneity

  - consider non-linear associations
  - align research designs more closely to the theories
  - use experimental designs

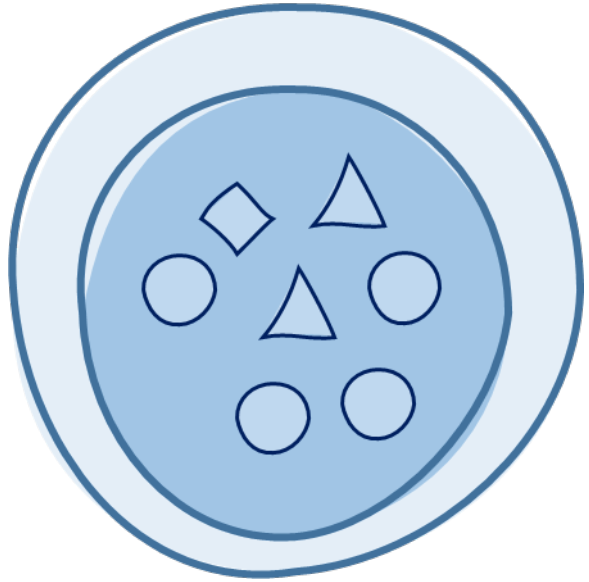
What are the reasons for differing findings?

Accumulate research on understudied dimensions!

Put theories to the test more rigorously!

? classroom composition recommendations/interventions ?

Thank you for your attention!



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